

Open weights are the publicly released learned parameters of a trained artificial intelligence model, principally its weights and biases. In an artificial neural network, weights are numerical values that determine how strongly inputs contribute to the network's calculations. Publishing these values allows others to download and use the model, while permission to modify, fine-tune or redistribute it depends on its license. The term is commonly applied to large language models and other generative artificial intelligence models.

Characteristics

During training, a neural network adjusts its parameters to improve its output. Weights determine the contribution of inputs to weighted sums, while biases are additional learned values. A model described as having seven billion parameters contains roughly seven billion learned values.

Model openness

Releasing model weights may form part of a broader effort to make an artificial intelligence model more transparent and reproducible. The Model Openness Framework evaluates whether components such as the model's parameters, source code, training data, evaluation results, intermediate checkpoints and technical documentation are publicly available under an open license. Under the framework, releasing only the final weights represents a lower degree of openness than releasing the materials needed to study, modify and reproduce the model.

Model cards

A model card is a document accompanying a trained machine learning model that describes a model's intended uses, limitations, training data and evaluation results. On Hugging Face, model cards are stored as Markdown README.md files with additional metadata in each model repository, helping users evaluate a model before downloading or deploying it.

Comparison with open-source AI

The Open Source Initiative distinguishes open weights from open-source artificial intelligence when the materials needed to study and modify the model's development process are not also available.

Benefits and concerns

Open weights can support local deployment, customization, independent research and competition without requiring access through the original developer's API. However, widely distributed weights generally cannot be withdrawn and may be modified to remove safeguards. A 2024 report by the National Telecommunications and Information Administration examined these benefits and risks and recommended continued monitoring rather than immediate restrictions.

Professors Peter Henderson and Mark Lemley argue that open-weight license terms are largely unenforceable because they presuppose intellectual property rights in ML model weights that do not exist. Whereas computer programs are typically copyrightable because they are "derived from a human-written copyrightable expression of an idea", model weights are generated automatically by a machine learning algorithm. The functional nature of ML model weights likely precludes copyright protection as well. A whitepaper published by the Intellectual Property Owners Association in 2020 proposes a sui generis right in trained AI models.

Open-weight large language models

The following is a list of notable large language models and model families for which at least one version has been released with publicly accessible weights.

See also

Comparison of free and open-source software licenses

Creative Commons license

Foundation model

Lists of open-source artificial intelligence software

List of proprietary artificial intelligence software

List of open-source software organizations

Open data

Source-available software

References

External links

AI openness: A primer for policymakers – OECD.AI

On the Societal Impact of Open Foundation Models